

- a. The Polynomial regression model has a longer training time; The KNN model takes little time to train as it only stores the data points whereas the Polynomial Regression model has to learn the parameters of a polynomial function. Its time complexity also increases with the degree of the polynomial
- b. The prediction time for KNN is higher; the model must calculate the distance between the test data points and all the training data points to find the nearest neighbors. This makes the prediction time of the KNN model highly dependent on the size of the test data, hence affecting prediction time.
- c. The KNN model would have a smaller file size, hence would be preferred. This is because the KNN model only needs to store the training data and the information about the number of nearest neighbors used for prediction. Moreover, the amount of file size needed to store this information is proportional to the number of features in the data.